

SUBJECT: Alarms Code List

Models affected: MR14-25, MR16N, MR16-20HD (D849)

Reason: Technical update

Technical Description:

In this Service Bulletin you find an updated overview of the D849 alarm codes.

In the third column you will find **X** if the alarm is new, **XX** if it is old but the name has been modified (the code is unchanged) and a blank box if nothing has changed.

From page 7 you can find the **causes and troubleshooting of the new alarms**.

VCM module alarms – software version 2.40:

CODE	MASTER ALARM - NODE 1	
8	WATCHDOG	
19	LOGIC FAILURE #1	
199	OVERLOAD	
200	WRONG PARAMETER	
201	WRONG SLAVE VER.	
202	HM MISMATCH	
203	TILLER MISMATCH	
204	BATTERY LOW	
205	RESET ENCODER	
206	BATTERY OUT	
208	EEPROM KO	
209	PARAM RESTORE	
210	WRONG RAM MEM.	
211	PUMP INC. START	
213	ENCODER LOCKED 1	
214	ENCODER LOCKED 2	
215	OUT PORT PULL-UP	

CODE	SLAVE ALARM - NODE 33	
8	WATCHDOG	
19	LOGIC FAILURE #1	
185	DRV. SHRT. EVP1 (LOWER)	XX
186	DRV. SHRT. EVP2 (LIFT)	XX
187	DRV. SHRT. EVP3 (REVERSE TROLLEY)	XX
188	DRV. SHRT. EVP4 (FORWARD TROLLEY)	XX
189	DRV. SHRT. EVP5 (LEFT AUX VALVE)	XX
190	DRV. SHRT. EVP6 (RIGHT AUX VALVE)	XX
191	DRV. SHRT. EVP7 (TILT/SIDE SHIFT SELECT VALVE)	XX
192	DRV. SHRT. EVP8 (5°WAY SELECT VALVE)	XX
193	DRV. OPEN EVP1 (LOWER)	X
194	DRV. OPEN EVP2 (LIFT)	X
195	DRV. OPEN EVP3 (REVERSE TROLLEY)	X
196	DRV. OPEN EVP4 (FORWARD TROLLEY)	X
197	DRV. OPEN EVP5 (LEFT AUX VALVE)	X
198	DRV. OPEN EVP6 (RIGHT AUX VALVE)	X
199	OUT1/2 COIL SH.(LIFT/LOWER)	XX

INTERCHANGEABILITY:	
PARTS STOCK:	Available from the EPO, Nijmegen, The Netherlands
EFFECTIVE SERIAL NUMBER:	

CODE	MASTER ALARM - NODE 1	
217	ANALOG INPUT	
218	RESET ENCODER KO	
219	VALVE ENABLE	
222	LIFT/LOW LEVER	
223	NO CAN MSG. SLAVE VCM	XX
224	WAITING FOR NODE	
225	CONTROLLER MISM.	
226	PUMP IN ALARM	
227	NO CAN MSG. PUMP	XX
228	NO CAN MSG. JOYSTICK	XX
229	SDO TRAC.	
230	INTERLOCK FORK	
231	CARRIAGE LEVER	
232	TILT LEVER	
233	SIDE SHIFT LEVER	
234	WAITING PCF	
235	WRONG CONFIG	
236	NO CONFIG BACKUP	
237	NO CONFIG MASTER	
238	PCF TIMEOUT	
239	REACH SENS OUT R	
240	REACH ACQ. SENS.	
241	M/S PAR CHK MISM	
242	PARAM TRANSFER	
243	PRESELECTOR	
244	CHECK UP	
245	SEAT BELT	XX
246	NO CAN MSG. TRACTION	XX
247	NO CAN MSG. EPS	XX
248	NO CAN MSG. DISPLAY	XX
249	CAN BUS DISPLAY	
250	CANTILLER	
251	TROLLEY BLOCKED	

CODE	SLAVE ALARM - NODE 33	
200	OUT3/4 COIL SH. (REV/FW TROLLEY)	XX
201	OUT5/6 COIL SH. (LEFT/RIGHTAUX VALVE)	XX
202	OUT7/8 COIL SH. (TILT/SIDESHIFT-5° WAY SELECT VALVE)	XX
203	LASER COIL SH.	
204	BAT OUT COIL SH.	
205	LOAD BRK COIL SH	
206	ALARM COIL SH.	
207	WATCH DOG MASTER	
208	EEPROM KO	
209	PARAM RESTORE	
210	WRONG RAM MEM.	
211	PEV DRV. OPEN	
212	PEV DRV. SHORT.	
213	VALVE MISM. OUT	
214	DRV. OPEN BRAKE	
215	OUT PORT PULL-UP	
217	ANALOG INPUT	
218	IN. MISM. D	
219	IN. MISM. A/E	
220	DRV. OPEN EV1	X
221	DRV. OPEN EV2(BACK UP ALARM)	X
222	DRV. OPEN EV3 (LASER)	X
223	NO CAN MSG. TRACTION	XX
224	NO CAN MSG. MASTER VCM	XX
225	CONTROLLER MISM.	
227	NO CAN MSG. PUMP	XX
228	NO CAN MSG. JOYSTICK	XX
229	DRV. OPEN EVP7 (TILT/SIDE SHIFT SELECT VALVE)	X
230	DRV. OPEN EVP8 (5°WAY SELECT VALVE)	X
235	DRV. SHRT. EV1	
236	DRV. SHRT. EV2 (BACK UP ALARM)	XX
237	DRV. SHRT BRAKE	
238	DRV. SHRT. EV3 (LASER)	XX
239	COIL OPEN EV1	
240	COIL OPEN EV2 (BACK UP ALARM)	XX
241	COIL OPEN BRAKE	
242	COIL OPEN EV3 (LASER)	XX
244	COIL OPEN EVP1 (LOWER)	XX
245	COIL OPEN EVP2 (LIFT)	XX
246	COIL OPEN EVP3 (REVERSE TROLLEY)	XX
247	COIL OPEN EVP4 (FORWARD TROLLEY)	XX
248	COIL OPEN EVP5 (LEFT AUX VALVE)	XX
249	COIL OPEN EVP6 (RIGHT AUX VALVE)	XX
250	COIL OPEN EVP7 (TILT/SIDE SHIFT SELECT VALVE)	XX
251	COIL OPEN EVP8 (5°WAY SELECT VALVE)	XX

ACE2 Traction module alarms – software version 2.15:

CODE	MASTER ALARM - NODE 2	
8	WATCHDOG	
17	LOGIC FAILURE #3	
18	LOGIC FAILURE #2	
19	LOGIC FAILURE #1	
30	VMN LOW	
31	VMN HIGH	
37	CONTACTOR CLOSED	
38	CONTACTOR OPEN	
53	STBY I HIGH	
60	CAPACITOR CHARGE	
62	TH. PROTECTION	
65	MOTOR TEMPERAT.	
66	BATTERY LOW	
74	DRIVER SHORTED	
75	CONTACTOR DRIVER	
78	VACC NOT OK	
79	INCORRECT START TRACT	
80	FORW + BACK	
82	ENCODER ERROR	
199	INCORRECT START PEDAL	
201	WRONG SLAVE VER.	
202	VDC LINK OVERV.	
203	TILLER MISMATCH	
204	BRAKE RUN OUT	
205	STOP RQ.FROM EPS	
206	INIT VMN HIGH	
207	INIT VMN LOW	
208	EEPROM KO	
209	PARAM RESTORE	
210	WRONG RAM MEM.	
211	STALL ROTOR	
212	WAITING SLAVE	
213	HARWARE FAULT	
214	POT ACCEL MISM.	
215	POT BRAKE MISM.	
218	SENS MOT TEMP KO	
219	PEV NOT OK	
220	VKEY OFF SHORTED	
223	MC COIL SHOR.	
224	WAITING FOR NODE	

CODE	SLAVE ALARM - NODE 34	
8	WATCHDOG	
17	LOGIC FAILURE #3	
18	LOGIC FAILURE #2	
19	LOGIC FAILURE #1	
30	VMN LOW	
31	VMN HIGH	
37	CONTACTOR CLOSED	
38	CONTACTOR OPEN	
53	STBY I HIGH	
60	CAPACITOR CHARGE	
62	TH. PROTECTION	
65	MOTOR TEMPERAT.	
66	BATTERY LOW	
74	DRIVER SHORTED	
75	CONTACTOR DRIVER	
78	VACC NOT OK	
79	INCORRECT START TRACT	
80	FORW + BACK	
82	ENCODER ERROR	
199	INCORECT START PEDAL	
201	WRONG SLAVE VER.	
202	VDC LINK OVERV.	
203	TILLER MISMATCH	
204	BRAKE RUN OUT	
205	STOP RQ.FROM EPS	
206	INIT VMN HIGH	
207	INIT VMN LOW	
208	EEPROM KO	
209	PARAM RESTORE	
210	WRONG RAM MEM.	
211	STALL ROTOR	
212	WAITING SLAVE	
213	HARWARE FAULT	
214	POT ACCEL MISM.	
215	POT BRAKE MISM.	
218	SENS MOT TEMP KO	
219	PEV NOT OK	
220	VKEY OFF SHORTED	
223	MC COIL SHOR.	
224	WAITING FOR NODE	

CODE	MASTER ALARM - NODE 2	
226	VACC OUT RANGE	
228	TILLER OPEN	
229	OUTPUT MISMATCH	
230	WD SYNCRO	
231	NO SLAVE ENABLE	
232	DIR SEL OUT RANG	
233	POWER MOS SHORT	
235	COIL SHOR. EB.	
236	CURRENT GAIN	
237	ANALOG INPUT	
239	CONTROLLER MISM.	
241	M/S PAR CHK MISM	X
242	PARAM TRANSFER	X
243	THROTTLE PROG.	
245	IQ MISMATCHED	
246	AUX DRIV.OPEN	
247	DATA ACQUISITION	
248	NO CAN MSG.	
250	THERMIC SENS. KO	
251	WRONG SET BAT.	
252	WRONG ZERO	
253	FIELD ORIENT. KO	
254	AUX DRIV.SHRT.	

CODE	SLAVE ALARM - NODE 34	
226	VACC OUT RANGE	
228	TILLER OPEN	
229	OUTPUT MISMATCH	
230	WD SYNCRO	
231	NO SLAVE ENABLE	
232	DIR SEL OUT RANG	
233	POWER MOS SHORT	
235	COIL SHOR. EB.	
236	CURRENT GAIN	
237	ANALOG INPUT	
239	CONTROLLER MISM.	
241	INPUT MISMATCH	
242	SP MISMATCH	
243	THROTTLE PROG.	
245	IQ MISMATCHED	
246	AUX DRIV.OPEN	
247	DATA ACQUISITION	
248	NO CAN MSG.	
250	THERMIC SENS. KO	
251	WRONG SET BAT.	
252	WRONG ZERO	
253	FIELD ORIENT. KO	
254	AUX DRIV.SHRT.	

ACE2 Pump module alarms – software version 2.15:

CODE	MASTER ALARM - NODE 5	
8	WATCHDOG	
17	LOGIC FAILURE #3	
18	LOGIC FAILURE #2	
19	LOGIC FAILURE #1	
30	VMN LOW	
31	VMN HIGH	
37	CONTACTOR CLOSED	
38	CONTACTOR OPEN	
53	STBY I HIGH	
60	CAPACITOR CHARGE	
62	TH. PROTECTION	
65	MOTOR TEMPERAT.	
74	DRIVER SHORTED	
75	CONTACTOR DRIVER	

CODE	SLAVE ALARM - NODE 37	
8	WATCHDOG	
17	LOGIC FAILURE #3	
18	LOGIC FAILURE #2	
19	LOGIC FAILURE #1	
30	VMN LOW	
31	VMN HIGH	
37	CONTACTOR CLOSED	
38	CONTACTOR OPEN	
53	STBY I HIGH	
60	CAPACITOR CHARGE	
62	TH. PROTECTION	
65	MOTOR TEMPERAT.	
74	DRIVER SHORTED	
75	CONTACTOR DRIVER	

CODE	MASTER ALARM - NODE 5	
82	ENCODER ERROR	
201	WRONG SLAVE VER.	
202	VDC LINK OVERV.	
204	BRAKE RUN OUT	
206	INIT VMN HIGH	
207	INIT VMN LOW	
208	EEPROM KO	
209	PARAM RESTORE	
210	WRONG RAM MEM.	
211	STALL ROTOR	
212	WAITING SLAVE	
213	HARDWARE FAULT	
218	SENS MOT TEMP KO	
219	PEV NOT OK	
220	VKEY OFF SHORTED	
223	MC COIL SHOR.	
224	WAITING FOR NODE	
228	TILLER OPEN	
229	OUTPUT MISMATCH	
230	WD SYNCRO	
231	NO SLAVE ENABLE	
233	POWER MOS SHORT	
235	COIL SHOR. EB.	
236	CURRENT GAIN	
237	ANALOG INPUT	
239	CONTROLLER MISM.	
241	M/S PAR CHK MISM	X
242	PARAM TRANSFER	X
245	IQ MISMATCHED	
246	AUX DRIV.OPEN	
247	DATA ACQUISITION	
248	NO CAN MSG.	
250	THERMIC SENS. KO	
251	WRONG SET BAT.	
252	WRONG ZERO	
253	FIELD ORIENT. KO	
254	AUX DRIV.SHRT.	

CODE	SLAVE ALARM - NODE 37	
82	ENCODER ERROR	
201	WRONG SLAVE VER.	
202	VDC LINK OVERV.	
204	BRAKE RUN OUT	
206	INIT VMN HIGH	
207	INIT VMN LOW	
208	EEPROM KO	
209	PARAM RESTORE	
210	WRONG RAM MEM.	
211	STALL ROTOR	
212	WAITING SLAVE	
213	HARDWARE FAULT	
218	SENS MOT TEMP KO	
219	PEV NOT OK	
220	VKEY OFF SHORTED	
223	MC COIL SHOR.	
224	WAITING FOR NODE	
228	TILLER OPEN	
229	OUTPUT MISMATCH	
230	WD SYNCRO	
231	NO SLAVE ENABLE	
233	POWER MOS SHORT	
235	COIL SHOR. EB.	
236	CURRENT GAIN	
237	ANALOG INPUT	
239	CONTROLLER MISM.	
241	INPUT MISMATCH	
242	SP MISMATCH	
245	IQ MISMATCHED	
246	AUX DRIV.OPEN	
247	DATA ACQUISITION	
248	NO CAN MSG.	
250	THERMIC SENS. KO	
251	WRONG SET BAT.	
252	WRONG ZERO	
253	FIELD ORIENT. KO	
254	AUX DRIV.SHRT.	

EPS module alarms – software version 6.21:

CODE	MASTER ALARM - NODE 6	
8	WATCHDOG	
13	EEPROM KO	
16	LOGIC FAILURE #4	
17	LOGIC FAILURE #3	
32	VMN NOT OK	
48	MAIN CONT. OPEN	
53	STBY I HIGH	
60	CAPACITOR CHARGE	
61	HIGH TEMPERATURE	
65	MOTOR TEMPERAT.	
70	HIGH CURRENT	
71	POWER FAILURE #3	
72	POWER FAILURE #2	
73	POWER FAILURE #1	
84	STEER SENSOR KO	
85	STEER HAZARD	
202	ACQ ABORTED	
203	ENC PULSES ACQ	
206	CAN BUS WARN	
207	SP MISMATCH	
208	OUT MISMATCH	
209	W.D. SYNCRO	
210	WRONG SLAVE VER.	
212	WRONG RAM MEM.	
213	PARAM RESTORE	
214	SP JERK	
215	CAN BUS KO M/S	
219	STEPPER MOT MISM	
220	MOTOR LOCKED	
221	M/S PAR CHK MISM	
225	CURRENT GAIN	
228	POSITION ERROR	
237	SLAVE ALARM	XX
238	EPS NOT ALIGNED	
240	KEYOFF	
241	FB SENSOR LOCKED	
242	Q LINE SENSOR KO	
243	D LINE SENSOR KO	
244	PARAM TRANSFER	
245	DATA ACQUISITION	
247	CAN BUS KO	
250	INPUT MISMATCH	
251	INIT VMN NOT OK	

CODE	SLAVE ALARM - NODE 38	
8	WATCHDOG	
13	EEPROM KO	
16	LOGIC FAILURE #4	
17	LOGIC FAILURE #3	
32	VMN NOT OK	
48	MAIN CONT. OPEN	
53	STBY I HIGH	
60	CAPACITOR CHARGE	
61	HIGH TEMPERATURE	
65	MOTOR TEMPERAT.	
70	HIGH CURRENT	
71	POWER FAILURE #3	
72	POWER FAILURE #2	
73	POWER FAILURE #1	
84	STEER SENSOR KO	
85	STEER HAZARD	
202	ACQ ABORTED	
203	ENC PULSES ACQ	
206	CAN BUS WARN	
207	SP MISMATCH	
208	OUT MISMATCH	
209	W.D. SYNCRO	
210	WRONG SLAVE VER.	
212	WRONG RAM MEM.	
213	PARAM RESTORE	
214	SP JERK	
215	CAN BUS KO M/S	
219	STEPPER MOT MISM	
220	MOTOR LOCKED	
221	M/S PAR CHK MISM	
225	CURRENT GAIN	
228	POSITION ERROR	
237	WAITING MASTER	XX
238	EPS NOT ALIGNED	
240	KEYOFF	
241	FB SENSOR LOCKED	
242	Q LINE SENSOR KO	
243	D LINE SENSOR KO	
244	PARAM TRANSFER	
245	DATA ACQUISITION	
247	CAN BUS KO	
250	INPUT MISMATCH	
251	INIT VMN NOT OK	

Causes and troubleshooting of new alarms:

In the following table you find the causes and troubleshooting of the new alarms.

VCM module alarms – software version 2.40:

CODE	SLAVE ALARM - NODE 33	CAUSE	TROUBLESHOOTING
193	DRV. OPEN EVP1 (LOWER)	VCM is not able to drive EVP1 (signal lowering the fork)	This type of fault is not related to external components; replace the logic controller.
194	DRV. OPEN EVP2 (LIFT)	VCM is not able to drive EVP2 (signal lifting the fork)	This type of fault is not related to external components; replace the logic controller.
195	DRV. OPEN EVP3 (REVERSE TROLLEY)	VCM is not able to drive EVP3 (trolley out)	This type of fault is not related to external components; replace the logic controller.
196	DRV. OPEN EVP4 (FORWARD TROLLEY)	VCM is not able to drive EVP4 (trolley in)	This type of fault is not related to external components; replace the logic controller.
197	DRV. OPEN EVP5 (LEFT AUX VALVE)	VCM is not able to drive EVP5 (negative tilt and left side shift)	This type of fault is not related to external components; replace the logic controller.
198	DRV. OPEN EVP6 (RIGHT AUX VALVE)	VCM is not able to drive EVP6 (positive tilt and right side shift)	This type of fault is not related to external components; replace the logic controller.
220	DRV. OPEN EV1	VCM is not able to drive EV1 (BLUE SPOT LIGHT)	This type of fault is not related to external components; replace the logic controller.
221	DRV. OPEN EV2 (BACK UP ALARM)	VCM is not able to drive EV2	This type of fault is not related to external components; replace the logic controller.
222	DRV. OPEN EV3 (LASER)	VCM is not able to drive EV3	This type of fault is not related to external components; replace the logic controller.
229	DRV. OPEN EVP7 (TILT/SIDE SHIFT SELECT VALVE)	VCM is not able to drive EVP7	This type of fault is not related to external components; replace the logic controller.
230	DRV. OPEN EVP8 (5°WAY SELECT VALVE)	VCM is not able to drive EVP8	This type of fault is not related to external components; replace the logic controller.

ACE2 Traction module alarms – software version 2.15:

CODE	MASTER ALARM - NODE 2	CAUSE	TROUBLESHOOTING
241	M/S PAR CHK MISM	Parameters are saved both in the master Eeprom and in the slave Eeprom. The two non-volatile memories must contain the same parameter values and they are compared periodically. If a difference is found, this alarm is shown. This alarm does not inhibit machine operation and default parameters are used.	Try to save the parameters again. If the fault reappears when the key switch is re-cycled, replace the controller.
242	PARAM TRANSFER	Parameters are saved both in the master Eeprom and in the slave Eeprom. The two non-volatile memories must contain the same parameter values and they are compared periodically. If the master is not able to transfer the parameters to the slave, this alarm is shown.	Try to save the parameters again. If the fault reappears when the key switch is re-cycled, replace the controller.

ACE2 Pump module alarms – software version 2.15:

CODE	MASTER ALARM - NODE 5	CAUSE	TROUBLESHOOTING
241	M/S PAR CHK MISM	Parameters are saved both in the master Eeprom and in the slave Eeprom. The two non-volatile memories must contain the same parameter values and they are compared periodically. If a difference is found, this alarm is shown. This alarm does not inhibit machine operation and default parameters are used.	Try to save the parameters again. If the fault reappears when the key switch is re-cycled, replace the controller.
242	PARAM TRANSFER	Parameters are saved both in the master Eeprom and in the slave Eeprom. The two non-volatile memories must contain the same parameter values and they are compared periodically. If the master is not able to transfer the parameters to the slave, this alarm is shown.	Try to save the parameters again. If the fault reappears when the key switch is re-cycled, replace the controller.